

Natural & Whole Numbers

Number World Study Notes

Counting numbers, whole numbers, and the number line

Numbers are the building blocks of mathematics. Before we explore fractions, negatives, or primes, we need to understand the simplest and most familiar numbers of all: the ones we use for counting. This chapter introduces natural numbers and whole numbers, and shows how they sit on the number line.

Key Definitions

Natural Numbers (N)

The counting numbers 1, 2, 3, 4, 5, ... used to count objects. $N = \{1, 2, 3, 4, \dots\}$. There is no largest natural number — the list goes on forever.

Whole Numbers (W)

All natural numbers together with 0. $W = \{0, 1, 2, 3, 4, \dots\}$. Every whole number is either 0 or a natural number.

Number Line

A straight line on which numbers are marked at equal distances apart, increasing from left to right. It gives a picture of how numbers are ordered.

Successor & Predecessor

The successor of a number n is $n + 1$ (the number right after it). The predecessor of a number n is $n - 1$ (the number right before it).

Key Facts & Rules

- Every natural number is a whole number, but 0 is a whole number that is NOT a natural number.
- The smallest natural number is 1. The smallest whole number is 0.
- There is no largest natural number or whole number — both sets are infinite.
- On the number line, whole numbers are placed at equal intervals starting from 0 and moving right.
- The predecessor of 0 does not exist among whole numbers (whole numbers cannot be negative).
- Between any two whole numbers that differ by more than 1, there are other whole numbers lying between them.

Worked Examples

Example 1: Find the successor of 999.

Successor = given number + 1.

$$999 + 1 = 1000.$$

Answer: 1000

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Example 2: Find the predecessor of 100.

Predecessor = given number - 1.

$$100 - 1 = 99.$$

Answer: 99

Example 3: Which of these numbers are whole numbers but NOT natural numbers: -3, 0, 5, 2.5?

-3 is negative, so it is not a whole number.

0 is a whole number, but not a natural number.

5 is both a natural number and a whole number.

2.5 is not a whole number (it is not a counting number).

Answer: Only 0

Example 4: List all whole numbers between 3 and 9.

'Between' means strictly greater than 3 and strictly less than 9.

These are 4, 5, 6, 7, 8.

Answer: 4, 5, 6, 7, 8

Practice Questions

- 1 Write the next three natural numbers after 47.
- 2 Is 0 a natural number? Justify your answer.
- 3 Find the predecessor of 1. Does it exist in whole numbers?
- 4 List all whole numbers between 10 and 16.
- 5 State True or False: 'Every whole number is a natural number.' Explain.

Answer Key

- 1 48, 49, 50.
- 2 No. Natural numbers start counting from 1; 0 is a whole number but not a natural number.
- 3 0. Yes, it exists since 0 is a whole number (but it has no predecessor itself).
- 4 11, 12, 13, 14, 15.
- 5 False — 0 is a whole number but is not a natural number, so it is a counterexample.